

Paper Manager pack — SigLIP (Sep 14)

Paper: Sigmoid Loss for Language Image Pre-Training (SigLIP)

arXiv: <https://arxiv.org/abs/2303.15343> · ICCV 2023 Oral

PDF: </workspace/papers/siglip-2303.15343.pdf>

Code: https://github.com/google-research/big_vision

Session: Sep 14 · due 9:30 AM · **You = Archaeologist (seat 1)**

Type: VLA Architectures

Sources: Summarizer + Senior Research (</workspace/siglip-quiet/DELIVERABLE.md>)

1. Paper summary

SigLIP replaces CLIP’s batch-softmax InfoNCE with a **pairwise sigmoid** loss over all image–text pairs (no global normalization across the batch). A chunked/distributed implementation scales to huge batches on few chips.

- **SigLiT** (locked image): up to **84.5%** ImageNet zero-shot on 4×TPUv4 in ~2 days.
- Batch gains **saturate ~32k**; larger (e.g. 307k) can hurt.
- Public **SigLIP So400M @ 729 patches** → **83.2%** ImageNet — the efficiency point that became the open VLA vision tower.
- **Why this club cares:** SigLIP-So400m → **PaliGemma** → π_0 / $\pi_{0.5}$ / **OpenPI** → 2025 BEHAVIOR winners’ vision stack.

2. Keywords / model bits

Kind	Terms
Loss	Pairwise sigmoid, learnable temperature t + bias b (b≈-10 init), no softmax denom
Models	SigLiT, SigLIP, mSigLIP; ViT + text Transformer; SoViT-400M
Data / scale	WebLI; batch sweeps 512 → 1M; XM3600 multilingual

Systems	Chunked distributed loss; $\beta_2=0.95$; no WD on unlocked pretrained vision
Downstream	PaliGemma vision tower; OpenPI / $\pi_{0.5}$; BEHAVIOR'25

Loss (sketch): $\mathcal{L} = -\frac{1}{|B|} \sum_{i,j} \log \sigma(\frac{z_{ij}}{\tau} (x_i^{\top} y_j + b))$ with $(z_{ij} = \pm 1)$ for match/non-match.

3. Role packs

Stakeholder (18+3) — seat 3

- **Motivation:** CLIP-style pretrain is the eyes of modern VLAs; better loss/systems = better open baselines.
- **Problem:** Softmax LIP needs huge synchronized batches; costly and saturates poorly engineered.
- **Method:** Sigmoid pairwise + chunked impl; locked (SigLiT) and unlocked (SigLIP) recipes; So400M public checkpoint.
- **Findings:** Small-BS sigmoid $\gg$ softmax; $\sim 32k$ enough; So400M hits 83.2% INet; lineage feeds PaliGemma $\rightarrow$ OpenPI $\rightarrow$ BEHAVIOR.
- **Takeaway for 2026 BEHAVIOR:** Don't reinvent CLIP softmax — start from SigLIP/So400M (or SigLIP2) and differentiate on **action / data / System-2 / control**.

Scientific Reviewer (8+3) — seat 5

- **Strengths:** Clean loss+systems co-design; rare mega-batch study; honest saturation; actionable ablations (bias, β_2 , WD).
- **Weaknesses:** Proprietary WebLI; softmax baseline less systems-optimized; thin theory; limited dense/VLM-as-encoder eval at publication; compute-matched SOTA imperfect.
- **Actionable:** Public-data replication; report VLM/robotics transfer; publish exact So400M $\rightarrow$ PaliGemma config; isolate loss vs data when claiming “SigLIP helps VLA.”

Empiricist (11+3) — seat 4

H (Senior Research): At $BS \leq 1k$, sigmoid+learned bias trains faster/stabler retrieval than matched softmax CLIP; removing bias hurts early stability. Success = qualitative Fig.2 gap, **not** paper ImageNet %.

Priority	Experiment
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P0 E1	Sigmoid vs softmax on tiny paired set (BS≤1k)
P0 E2	Bias b=−10 ablation
P0 E3	Frozen SigLIP vs CLIP linear probe on household/robot stills
P1 E4	OpenPI/ $\pi_{0.5}$ encoder-swap **outline only**
Skip	Full WebLI-scale reproduce

Budget: E1/E2 ≈ 1×24–40GB overnight (6–24h); E3 cheap. Env stubs: /workspace/siglip-quiet/env_outline/ (open_clip_torch / timm).

Archaeologist (11+3) — ****YOU (seat 1)**** — HIGH PRIORITY

Influential priors that shaped SigLIP

Prior	How it shaped this paper
CLIP (Radford et al.)	Dual-encoder LIP setup SigLIP surgically replaces (softmax → sigmoid)
ALIGN	Large-scale noisy web image–text; scale narrative
LiT	Locked-image tuning path → **SigLiT**
InfoNCE / SimCLR	Contrastive softmax lineage being replaced
ViT / Scaling ViTs	Vision backbone family; SoViT shape-opt for So400M
WebLI / PaLI	Data + multilingual / VLM context
Sigmoid-in-classification (e.g. ReaL)	Motivates pairwise logistic over exclusive softmax
Related contemporaneous	FLIP, BASIC, CoCa, BLIP (competing LIP recipes)

Subsequent work that builds on SigLIP (clear line)

Later work	How it extends / uses SigLIP
PaLI-3	Continues Google VLM stack with SigLIP-family vision
PaliGemma (arXiv 2407.07726)	**SigLIP-So400m + Gemma-2B** — the critical bridge
π_0 (2410.24164)	Initializes from PaliGemma → robot VLA
$\pi_{0.5}$ / OpenPI	Same vision tower lineage in open robot stacks
SigLIP 2 (2502.14786)	Next-gen SigLIP (dense/multilingual improvements)
BEHAVIOR’25 1st place (2512.06951)	Diagram: SigLIP → PaliGemma → Action Expert; q≈0.26
HF Transformers SigLIP APIs	Mass adoption / drop-in weights

One-sentence lineage for your talk:

*CLIP/ALIGN/LiT + WebLI → **SigLIP** → PaliGemma → π_0 /OpenPI → $\pi_{0.5}$ → 2025 BEHAVIOR winners (and → SigLIP 2).*

Honest hedges: Exact OpenPI freeze-vs-unlock of SigLIP weights in every fork is not always documented; many “uses SigLIP” papers inherit weights **without** isolating loss vs data/architecture.

Visionary (11+3) — seat 2

- **2026 BEHAVIOR:** Treat SigLIP → PaliGemma → $\pi_{0.5}$ as default open eyes; try **SigLIP 2** dense/multilingual behind the **same** action expert; measure encoder swaps in **q-score** under fixed policy compute (not COCO R@1).
- Pairwise loss invites async/hard-negatives for small robot batches; NaFlex / resolution tricks for egocentric cams.
- Skip Discord/Drive.

4. Link to long-horizon control (club thread)

SigLIP is **perception**, not a planner. BEHAVIOR failures more often sit in **chunking / stages / recovery / data** than ImageNet-0 of the vision tower — but SigLIP-So400m is still the **default open eyes** those controllers see through. Don’t claim SigLIP “solves” motion planning; claim it **standardized the backbone** the planning/control stack sits on.

5. Slide Maker brief

Role	Timing	Focus
Stakeholder	18+3	Why sigmoid; So400M → PaliGemma → BEHAVIOR chain
Reviewer	8+3	Strengths + WebLI/repro asks
Empiricist	11+3	Small-BS sigmoid vs sigmoid; bias; probe
Archaeologist	**11+3**	**Prior table + subsequent timeline (HIGH)**
Visionary	11+3	2026 encoder-swap experiments

Image-first; notes in speaker notes; **PDF exports for Mac**. No Stakeholder redo in later roles.

Paths

Artifact	Path
This pack	`/workspace/siglip-quiet/KANTA_PACK.md`

Senior Research	`/workspace/siglip-quiet/DELIVERABLE.md`
Summarizer dump	`/workspace/papers/siglip-2303.15343-summary.md`
Env outline	`/workspace/siglip-quiet/env_outline/`